STUDYLENS – STUDENT LEARNING ANALYZER

A PROJECT REPORT

Submitted by

ASHVANTH M

(241501024)

in partial fulfilment of the End Semester Examination for the course

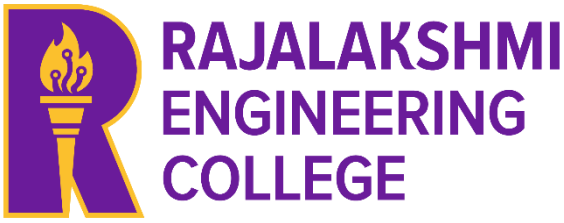
CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF TECHNOLOGY

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

OCTOBER 2026

RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "STUDYLENS – STUDENT LEARNING ANALYZER" is the bonafide work of ASHVANTH M (241501024), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester III, under my supervision.

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

DECLARATION

I, ASHVANTH M (241501024), hereby declare that the project report titled "STUDYLENS – STUDENT LEARNING ANALYZER", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports and dashboard — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date: 19 October 2026

SIGNATURE OF THE STUDENT
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ACKNOWLEDGEMENT

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ASHVANTH M

ABSTRACT

Teachers routinely maintain student details, subject and topic lists and assessment marks, and use these records to judge how well students are learning. When this information is kept in notebooks, spreadsheets or separate files, it becomes scattered, percentages have to be calculated repeatedly by hand, and identifying students who may need additional academic support requires a manual review of many records.

StudyLens – Student Learning Analyzer is a simple student learning analysis application developed on the Zoho Creator low-code platform to bring these activities into one place. The application is organised into six modules: Dashboard, Students, Subjects, Topics, Assessments and Learning Gap Analysis. Structured forms capture student, subject, topic and assessment data, and each topic is associated with a subject and a difficulty level of Easy, Medium or Hard. For every assessment record, the percentage is derived as (Marks Obtained ÷ Total Marks) × 100, and a simple threshold rule labels results below 50% as "Needs Improvement" and results of 50% or above as "Good Progress". Reports list the stored records, and the dashboard is intended to summarise the total number of students, subjects, topics and assessments together with a chart of assessment marks. The analysis is rule-based: the application does not use a trained artificial intelligence or machine learning model, and no external integrations are used.

Sixteen test cases covering data entry, percentage calculation, 50% boundary classification, rejection of invalid marks and dashboard totals have been executed and verified against the application logic. The project demonstrates how the forms, reports, Deluge workflows and dashboard features of Zoho Creator reduce the amount of conventional code needed for a practical academic record-keeping application, while showing that correct data modelling, validation and testing remain essential.

Keywords: *Low-Code Development; Zoho Creator; Student Assessment; Learning Gap Analysis; Rule-Based Classification; Educational Data Management*

PROJECT AT A GLANCE

Attribute	Details
Project Title	STUDYLENS – STUDENT LEARNING ANALYZER
Application Domain	Education – Student Assessment Management
Student Name & Register No.	ASHVANTH M – 241501024
Zoho Creator Edition	Zoho Creator Free / Education Trial Edition
Application Name (Link Name)	StudyLens (link name: studylens-student-learning-analyzer)
Published / Live URL	https://creatorapp.zoho.in/ashvanth.m.2024.aiml_ra/studylens-student-learning-analyzer
Demo Credentials (per role)	Teacher / Application Owner: ashvanth.m.2024.aiml@rajalakshmi.edu.in
Number of Forms / Subforms	4 forms (Students, Subjects, Topics, Assessments) / 0 subforms
Number of Reports (by type)	List: 5 (All Students, All Subjects, All Topics, All Assessments, Learning Gap Analysis) Kanban: 0 Calendar: 0 Chart: 1 (Assessment Marks Chart on Dashboard) Others: 0
Number of Pages / Dashboards	1 (Dashboard)
Number of Workflows	Form: 2 (On Validate Marks Validation & Percentage/Status Calculation) Schedule: 0 Approval: 0 Blueprint: 0 Batch: 0
Deluge Functions / Custom Actions	2 Deluge On Validate Scripts (DS-01 Marks Validation & DS-02 Percentage and Performance Status Calculation)
Integrations	None (not implemented in the current version)
Roles & Profiles Defined	Application Owner / Teacher (Default Administrator access; custom roles not required for single-teacher scope)
Environments Used	Development Environment (Single-environment Zoho Creator workspace)
Date of Production Deployment	October 2026

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
AI	Artificial Intelligence
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
ER	Entity–Relationship
FK	Foreign Key
FR	Functional Requirement
IEEE	Institute of Electrical and Electronics Engineers
ML	Machine Learning
NFR	Non-Functional Requirement
RBAC	Role-Based Access Control
REST	Representational State Transfer
UAT	User Acceptance Testing
UI	User Interface
Zia	Zoho’s AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

Assessment is a routine part of teaching. After each test, quiz or assignment, a teacher records the marks obtained by every student, compares them with the maximum marks of the assessment and uses the results to decide which students or topics need more attention.

In many classrooms these records are maintained in notebooks, spreadsheets or separate files for each subject or assessment. Student details, subject lists, topic lists and marks are therefore stored in different places. Converting marks into percentages, checking which students scored below an expected level and preparing a summary for a class all require repeated manual effort.

StudyLens – Student Learning Analyzer has been developed as an individual academic project for the course CB23F35 – Low-Code Application Development using Zoho Creator. It is a small-scale application intended for a teacher who wants to organise student assessment data in one place and quickly identify assessment records that may need follow-up. The main purpose of the project is to demonstrate how Zoho Creator forms, reports, calculated fields and dashboards can simplify student assessment management.

The sample records used in this report — students named Bala, Arjun, Sanjay, Akash and Karthik, and subjects such as Mathematics, Python Programming and Database Management — are fictional demonstration data. They do not represent real students or an official course curriculum.

1.2 Problem Statement

Teachers need a convenient way to maintain student details, organise subjects and topics, record assessment results and identify students who may need academic improvement. When these activities are performed manually or across separate files, record management and performance analysis require additional effort, percentages must be calculated repeatedly, and locating students who score below a defined threshold is inconvenient. A simple, centralised application is therefore required to

manage student assessment information and to present basic performance indicators in an understandable format. The specific pain points are:

- Student, subject, topic and assessment information is scattered across notebooks, spreadsheets or separate files.
- Percentages are calculated manually for every assessment, which is repetitive and prone to calculation errors.
- It is difficult to view all assessment records of a student, subject or topic together.
- Identifying low-scoring assessment records requires scanning many records by hand.
- There is no quick visual summary of the overall assessment data.

1.3 Objectives

- To create a centralised form for maintaining student information.
- To organise subjects and their corresponding learning topics.
- To record assessment marks for individual students.
- To calculate assessment percentages from marks obtained and total marks.
- To identify students who may need improvement using a simple threshold rule.
- To provide reports and a dashboard for viewing assessment information.
- To demonstrate the use of Zoho Creator's low-code development features in an educational application.

1.4 Scope of the Project

StudyLens focuses on basic student assessment management and learning performance analysis for a single teacher. The scope of the project includes:

- Student information management (name, email and course).
- Subject management and topic management, with a difficulty level for each topic.
- Assessment record management (total marks and marks obtained for a student, subject and topic).
- Percentage calculation and basic learning gap identification using a 50% threshold.
- Reports and dashboard summaries presented through a simple, understandable user interface.

The application is accessed through a web browser on the Zoho Creator platform by the teacher, who is also the application owner in the current version. Access through the Zoho Creator mobile app and a customer portal is not part of the current scope.

Out of scope. StudyLens does not automatically diagnose the reasons for poor academic performance. It does not use a trained machine learning model, predict future grades or provide personalised AI-generated teaching plans, and it does not integrate with any external system. Features that are not implemented are documented as future enhancements in Section 9.2.

1.5 Justification for a Low-Code Approach

A low-code platform lets developers build applications mainly by configuring visual models of data, user interfaces and logic, with hand-written code needed only for specific requirements [1]. Zoho Creator provides a visual form builder, automatically generated reports, a page builder for dashboards and a hosted cloud environment, with the Deluge scripting language available when custom logic is required [2].

StudyLens is well suited to this approach. Its data consist of structured records (students, subjects, topics and assessments), and its main requirements are data-entry forms, record lists, a simple percentage calculation, a threshold rule and a summary dashboard — all of which map directly onto standard Zoho Creator components. Building the same application in a conventional way would require a separately designed database, server-side code, user interface pages and hosting.

Using Zoho Creator therefore reduces the amount of conventional code to be written, removes the need to set up a server or database, and makes it easier to add or change fields later. A low-code platform does not remove the need for careful design: correct data modelling, validation of entered marks, testing and access control still have to be planned and verified by the developer.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance.

Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In a conventional approach, student information, subject details, topic lists and assessment marks are commonly maintained in separate spreadsheets or paper records. The teacher writes or types student details, keeps subject and topic lists in separate files, enters marks for each assessment in a different sheet, calculates percentages manually, and then scans the records to find students who scored below an expected level before preparing any summary by hand. Figure 2.1 shows this general as-is process. It describes the general limitations of manual record keeping and is not a claim about the exact process followed at any particular institution.

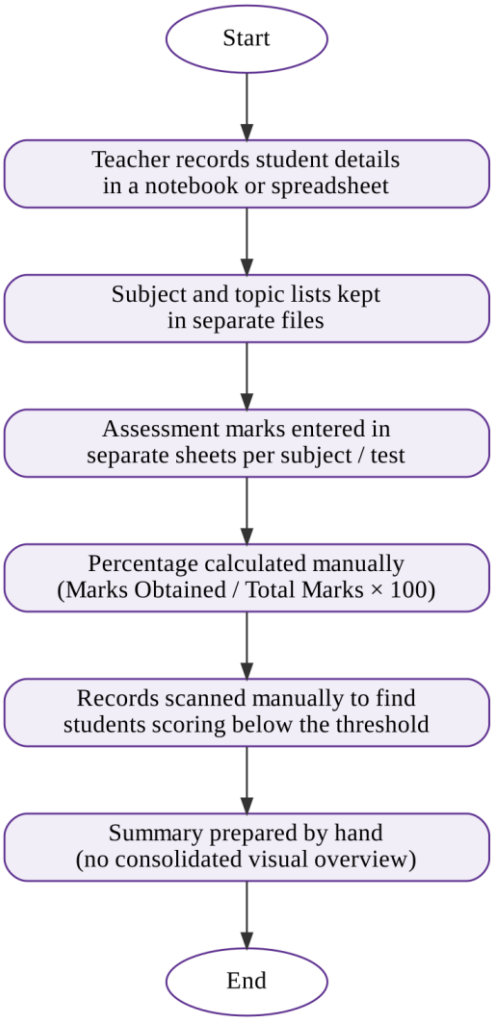


Figure 2.1: As-Is Process Flow of the Existing (Manual) System

2.2 Limitations of the Existing System

- **Scattered information:** student, subject, topic and mark records are kept in different places.
- **Repeated manual calculations:** percentages have to be calculated by hand for every assessment.
- **No combined view:** it is difficult to view all assessment records together.
- **Additional effort to find low scores:** identifying low-scoring students requires reviewing many records.
- **Limited visual summaries:** there is no ready overview of student performance.

2.3 Low-Code versus Traditional Development

Table 2.1 compares a conventional (pro-code) implementation of StudyLens with its low-code implementation on Zoho Creator. The criteria are adapted from feature-based comparisons of low-code development platforms [3]. Development effort is described qualitatively because the time taken was not formally measured.

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Longer: the database schema, server-side code, user interface pages and hosting must each be built and connected.	Shorter: forms, reports and the dashboard are configured visually; the actual effort for StudyLens was not formally measured.
Skills required	A programming language, a web framework, SQL and deployment skills.	Data modelling and Zoho Creator configuration; Deluge scripting only where custom logic is needed.
Hosting & infrastructure	A web server and database must be provisioned and maintained separately.	Hosted on the Zoho Creator cloud; no server set-up is required from the developer.
Mobile application	A responsive design or a separate mobile application must be developed.	Zoho Creator offers mobile access to applications; not tested for StudyLens in the current version.
Security & access control	User authentication and authorisation must be coded and tested.	Uses Zoho account sign-in and platform permission settings (custom roles are not configured in StudyLens).

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Maintenance & change	Each change requires code modification, testing and redeployment.	Fields, forms and reports can be modified in the visual builder with little or no code.
Customisation limits	Fully flexible user interface and logic.	Limited to the components, layouts and scripting features offered by the platform.
Cost	Developer time plus separate hosting costs.	Depends on the Zoho Creator plan used; no separate hosting cost.

2.4 Literature Survey

The sources reviewed fall into two groups. The first group consists of studies of low-code development platforms [1], [3], [4] and the official Zoho Creator documentation, which inform how StudyLens is built. The second group consists of educational research on classroom assessment [5] and learning analytics [6], which provides general background on why recorded assessment results are worth analysing. Table 2.2 summarises these sources. The educational sources are background only: StudyLens implements a simple percentage-threshold rule and does not implement the methods discussed in them.

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	A. C. Bock and U. Frank, 2021 [1]	Low-Code Platform	Describes low-code platforms as environments in which applications are built mainly by configuring models of data, user interfaces and processes, reducing hand-written code while tying the application to the concepts of the platform.	Explains why a form-and-report based application such as StudyLens suits a low-code platform, and the trade-offs involved.

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
2	A. Sahay et al., 2020 [3]	Supporting the Understanding and Comparison of Low-Code Development Platforms	Analyses a selection of low-code development platforms and proposes a set of features for understanding and comparing them.	Provides the basis for the comparison criteria used in Table 2.1.
3	D. Di Ruscio et al., 2022 [4]	Low-Code Development and Model-Driven Engineering: Two Sides of the Same Coin?	Discusses the relationship between low-code development and model-driven engineering, and the opportunities and challenges they share.	Helps explain that StudyLens forms act as models from which Zoho Creator generates the data store and user interface.
4	P. Black and D. Wiliam, 1998 [5]	Assessment and Classroom Learning	A review of research on formative assessment, concluding that using assessment information to adapt teaching can improve student learning.	Background for recording assessment results so that students needing attention can be identified (not implemented as a teaching method in StudyLens).
5	P. Long and G. Siemens, 2011 [6]	Penetrating the Fog: Analytics in Learning and Education	Describes learning analytics as the measurement, collection, analysis and reporting of data about learners and their contexts in order to improve learning.	Positions the percentage and threshold analysis of StudyLens as a basic form of reporting on learner data, not advanced analytics.
6	Zoho Corporation [2]	Zoho Creator Help Documentation	Official documentation of Zoho Creator forms, fields, reports, pages, workflows and user management.	Primary technical reference used to design and build StudyLens.

2.5 Emerging Trends in Low-Code Development

AI-assisted application building. Low-code platforms increasingly offer AI-assisted features, such as generating application structures or scripts from natural-language

descriptions. These features were not used to build StudyLens; all forms, reports and the dashboard were configured by the developer.

Citizen development. Low-code tools allow people with domain knowledge but limited programming experience to build simple applications. StudyLens reflects this trend: its structure (students, subjects, topics, assessments) mirrors how a teacher already thinks about assessment records, so it can be understood and modified without deep programming knowledge.

Convergence with model-driven engineering. Low-code development is closely related to model-driven engineering, in which models are the main development artefacts [4]. In StudyLens, the form definitions act as the model from which the platform generates storage and user interface.

Learning analytics. Educational institutions are increasingly interested in using learner data to support teaching [6]. StudyLens adopts only a very simple, transparent form of this idea — a percentage threshold — so that every result can be explained during review. More advanced analytics are listed as future enhancements.

2.6 Proposed System

The proposed system, StudyLens, is a centralised Zoho Creator application that contains forms to capture student, subject, topic and assessment information, reports to display the saved records, and a dashboard to summarise the available information. Assessment percentages are calculated using the formula:

$$\textbf{Percentage} = (\textbf{Marks Obtained} / \textbf{Total Marks}) \times 100$$

A simple rule then classifies each result: a percentage below 50% is labelled “Needs Improvement”, and a percentage of 50% or above is labelled “Good Progress”. The calculation is performed only when Total Marks is greater than zero, so that a zero or missing total cannot cause a division error. This rule is an indicative academic performance classification; it does not establish a student’s complete learning ability or diagnose the cause of a low score.

The proposed system addresses the limitations listed in Section 2.2 as follows: scattered information is replaced by one application with related forms; manual calculations are replaced by a consistent percentage calculation; the reports provide a combined view

of records; the Learning Gap Analysis view makes low-scoring records easy to locate; and the dashboard provides a visual summary.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Teacher (Primary User)	The person who uses StudyLens to manage academic records of a class.	Add and update students, subjects, topics and assessments; view reports, the Learning Gap Analysis and the dashboard.
Application Developer / Administrator	The student developer who builds and owns the Zoho Creator application.	Design and maintain forms, fields, reports and the dashboard; manage application settings.
Students (Indirect Stakeholders)	Learners whose assessment results are recorded in the application.	No direct access in the current version; they benefit when the teacher reviews records marked “Needs Improvement”.

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall allow the teacher to add, view and update student records with Student Name, Email and Course.	Teacher	High
FR-02	The system shall allow the teacher to add, view and update subjects.	Teacher	High
FR-03	The system shall allow the teacher to add topics, each associated with a subject and a difficulty level (Easy, Medium or Hard).	Teacher	Medium
FR-04	The system shall allow the teacher to record an assessment with Student, Subject, Topic, Total Marks and Marks Obtained.	Teacher	High
FR-05	The system shall reject an assessment where Total Marks is zero or missing, or Marks Obtained is negative or greater than Total Marks.	Teacher	High
FR-06	The system shall calculate Percentage = (Marks Obtained / Total Marks) × 100 for each valid assessment.	Teacher	High
FR-07	The system shall label an assessment “Needs Improvement” if the percentage is below 50% and “Good Progress” if it is 50% or above.	Teacher	High

FR ID	Requirement	Role(s)	Priority
FR-08	The system shall provide reports listing the records of each module.	Teacher	Medium
FR-09	The system shall provide a Learning Gap Analysis view showing Student, Subject, Topic, Marks Obtained, Total Marks, Percentage and Performance Status.	Teacher	High
FR-10	The system shall display a dashboard with the total number of students, subjects, topics and assessments and a chart of assessment marks.	Teacher	Medium

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Forms and reports shall open within an acceptable time for a class-sized data set (not formally measured in this project).
NFR-02	Security	Student data shall be accessible only to authorised Zoho account users of the application.
NFR-03	Usability	The interface shall use clear labels, a simple layout and a minimal number of fields per form.
NFR-04	Availability	The application shall be available wherever the Zoho Creator service and an internet connection are available.
NFR-05	Maintainability	Fields, forms and reports shall be modifiable through the Zoho Creator builder without rewriting the application.
NFR-06	Data Integrity	Invalid marks shall be rejected so that every stored percentage lies between 0% and 100%.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator (Free / Education Trial Edition)
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment) – used on the Assessments form On Validate event for marks validation and percentage/status calculation
Browser	A modern web browser such as Google Chrome or Microsoft Edge (latest version)
API testing tool	Not required (no API integration in the current version)
Other Zoho services	None used
Mobile	Not used in the current version
Client hardware	Desktop computer or laptop with keyboard, mouse and a reliable internet connection; no specialised hardware is required
Documentation	Microsoft Word or a compatible document editor for the project report

3.5 Use Case Model

Figure 3.1 shows the use case diagram of StudyLens. The teacher is the primary actor who manages records and views the analysis; the application developer/administrator

configures the application. Students are indirect stakeholders and do not interact with the application. The key use case is described in Table 3.5.

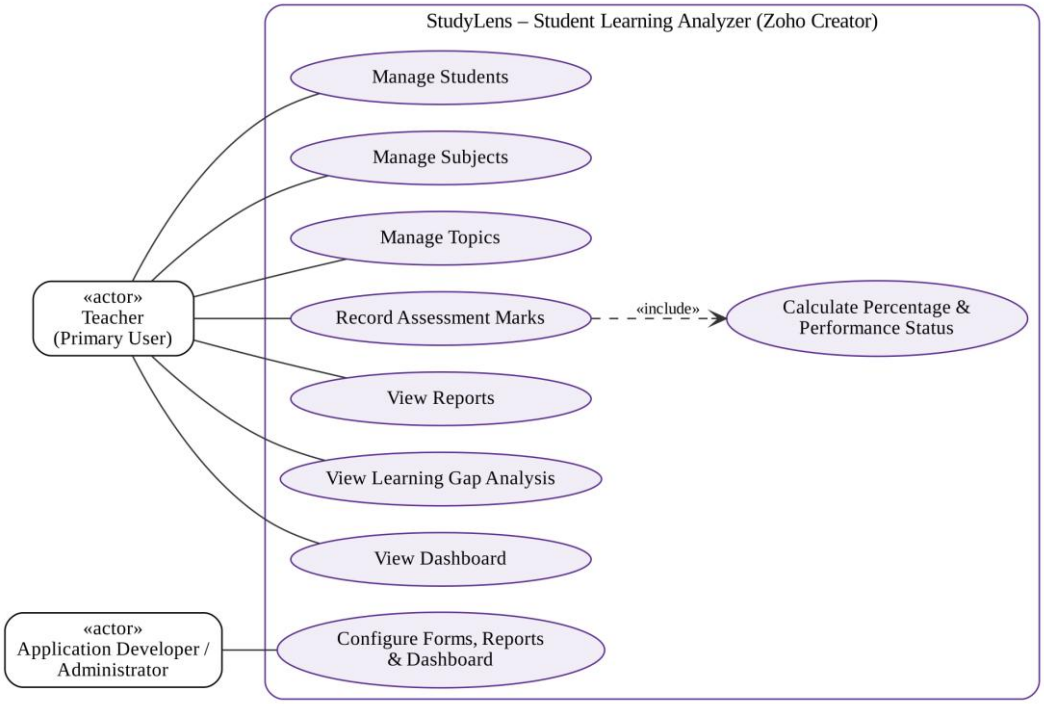


Figure 3.1: Use Case Diagram of StudyLens

Table 3.5: Use Case Description – Record Assessment and Review Performance Status

Use Case ID	UC-01
Use Case Name	Record Assessment and Review Performance Status
Actor(s)	Teacher
Pre-condition	The student, subject and topic records already exist in StudyLens.
Main Flow	1. The teacher opens the Assessments form. 2. The teacher selects the student, subject and topic. 3. The teacher enters Total Marks and Marks Obtained and submits the form. 4. The application validates the marks. 5. Percentage and Performance Status are derived. 6. The record appears in the reports, the Learning Gap Analysis and the dashboard.
Alternate Flow	If Total Marks is zero or missing, or Marks Obtained is negative or greater than Total Marks, an error message is shown and the record is not saved.
Post-condition	A valid assessment record is stored with its percentage and performance status.
Related FR	FR-04, FR-05, FR-06, FR-07, FR-09

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

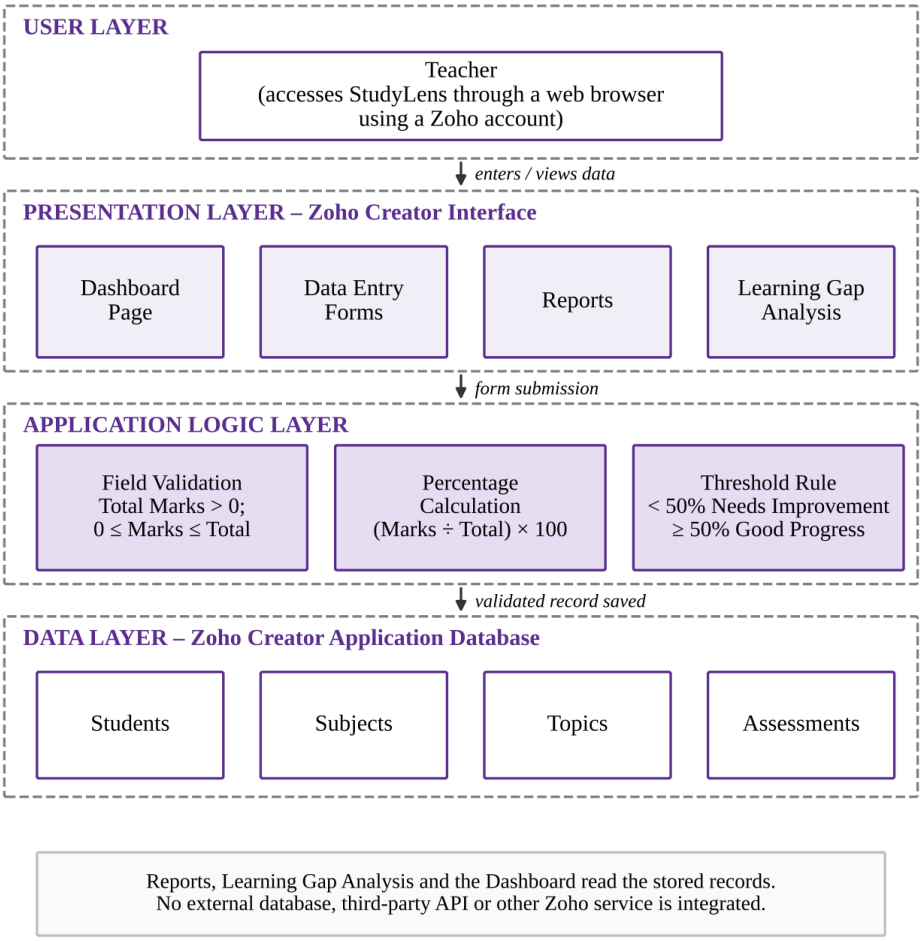


Figure 4.1: System Architecture of StudyLens

Figure 4.1 shows the layered architecture of StudyLens. The flow of information is: User → Zoho Creator Interface → Forms and Reports → Application Data → Dashboard and Learning Gap Analysis.

User layer. The teacher accesses StudyLens through a web browser after signing in with a Zoho account.

Presentation layer. The Zoho Creator interface presents the Dashboard page, the data-entry forms, the reports and the Learning Gap Analysis view. These are generated by the platform from the configured forms, reports and page.

Application logic layer. When an assessment is submitted, the entered marks are validated, the percentage is calculated and the 50% threshold rule is applied.

Data layer. Submitted records are stored in the built-in database of the Zoho Creator application, one record set per form (Students, Subjects, Topics and Assessments). The reports, the Learning Gap Analysis and the dashboard read these stored records. No external database or integration is used.

4.2 Data Model

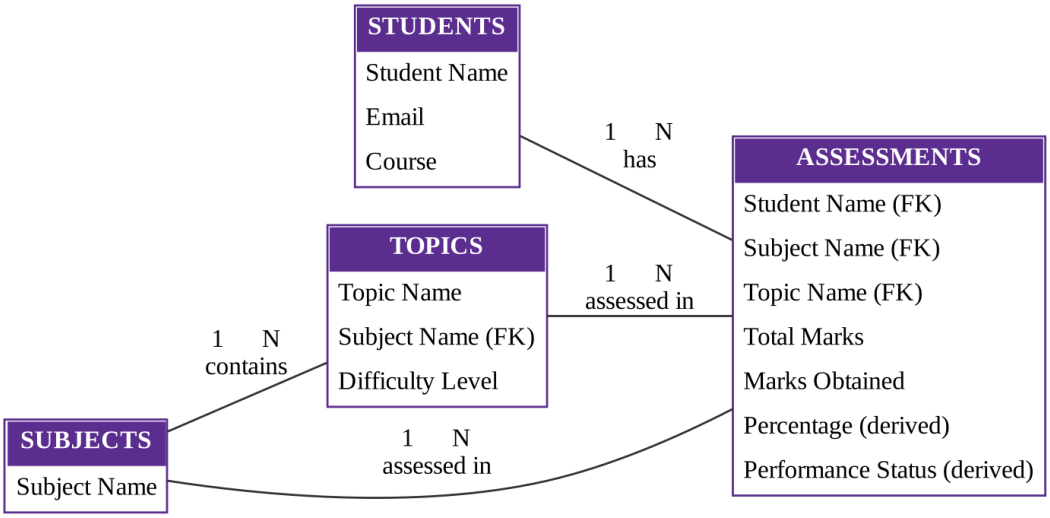


Figure 4.2: Entity–Relationship Diagram of StudyLens (Conceptual)

StudyLens has four entities, each implemented as a Zoho Creator form. A subject contains many topics, and each assessment record links one student with one subject and one topic. Percentage and Performance Status are derived attributes of an assessment. In Zoho Creator, such one-to-many relationships are realised through lookup fields. The relationships in Figure 4.2 and Table 4.1 represent the relational data model of the application.

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Students	Student Name, Email, Course	Assessments	One-to-Many	Lookup to Students in the Assessments form
Subjects	Subject Name	Topics	One-to-Many	Lookup to Subjects in the Topics form

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Subjects	Subject Name	Assessments	One-to-Many	Lookup to Subjects in the Assessments form
Topics	Topic Name, Subject Name, Difficulty Level	Assessments	One-to-Many	Lookup to Topics in the Assessments form
Assessments	Total Marks, Marks Obtained, Percentage, Performance Status	Students, Subjects, Topics	Many-to-One	Lookup fields (see above)

4.3 Form Design

Tables 4.2 to 4.5 give the field specification of the four StudyLens forms. The link names shown are the standard field link names generated by Zoho Creator from the field labels.

Table 4.2: Field Specification – Students Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Student Name	Student_Name	Single Line	Yes	—	Sample values: Bala, Arjun, Sanjay, Akash, Karthik
Email	Email	Email	Yes	Valid email format (checked by the field type)	Contact email of the student
Course	Course	Single Line	Yes	—	Programme / course of the student

Table 4.3: Field Specification – Subjects Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Subject Name	Subject_Name	Single Line	Yes	No duplicate values (recommended)	e.g., Mathematics, Python Programming, Database Management

Table 4.4: Field Specification – Topics Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Topic Name	Topic_Name	Single Line	Yes	—	Learning topic within a subject
Subject Name	Subject_Name	Lookup (Subjects)	Yes	Must select an existing subject	Associates the topic with a subject
Difficulty Level	Difficulty_Level	Drop Down	Yes	Choices: Easy / Medium / Hard	Indicates topic difficulty

Table 4.5: Field Specification – Assessments Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Student Name	Student_Name	Lookup (Students)	Yes	Must select an existing student	—
Subject Name	Subject_Name	Lookup (Subjects)	Yes	Must select an existing subject	—
Topic Name	Topic_Name	Lookup (Topics)	Yes	Topic should belong to the selected subject	—
Total Marks	Total_Marks	Number	Yes	Must be greater than 0	Maximum marks of the assessment
Marks Obtained	Marks_Obtained	Decimal	Yes	Must be ≥ 0 and ≤ Total Marks	Marks scored by the student
Percentage	Percentage	Formula / Decimal	—	(Marks_Obtained / Total_Marks) × 100, only when Total Marks > 0	Derived field
Performance Status	Performance_Status	Formula / Single Line	—	“Needs Improvement” if Percentage < 50; otherwise “Good Progress”	Derived field

4.4 Process and Workflow Design

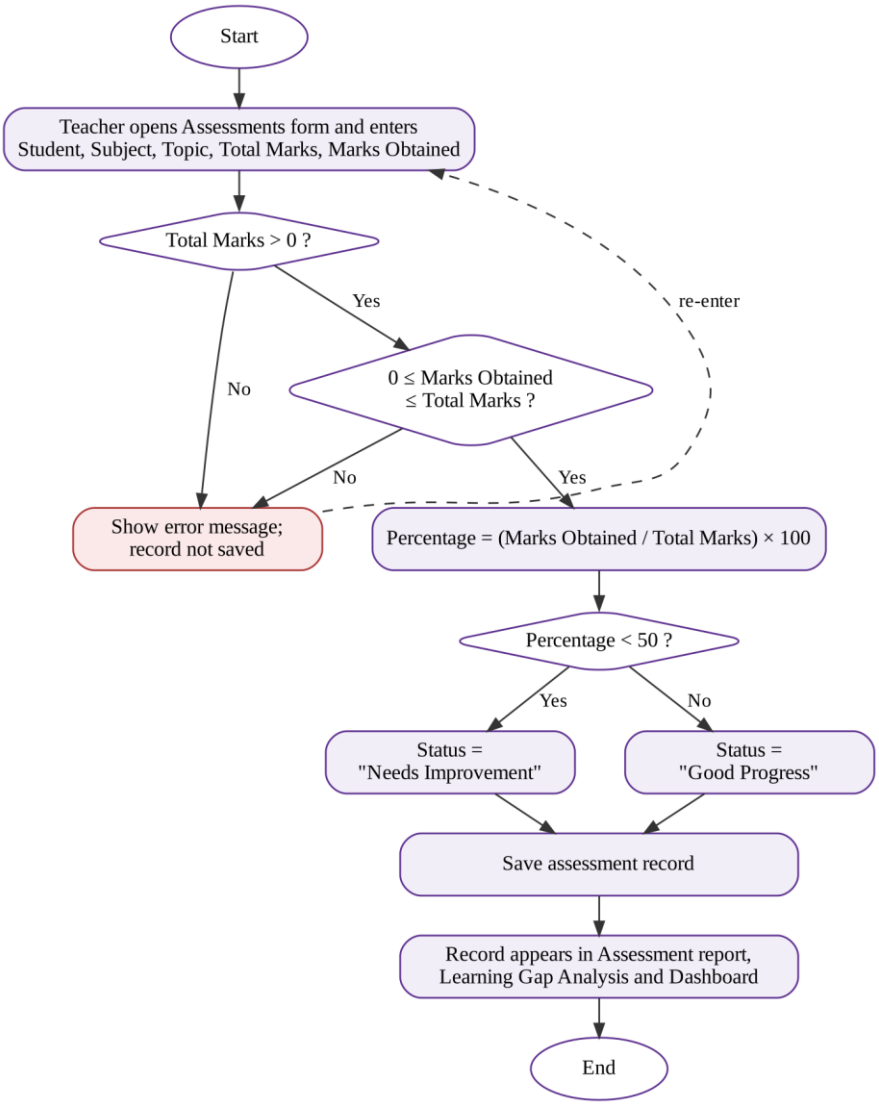
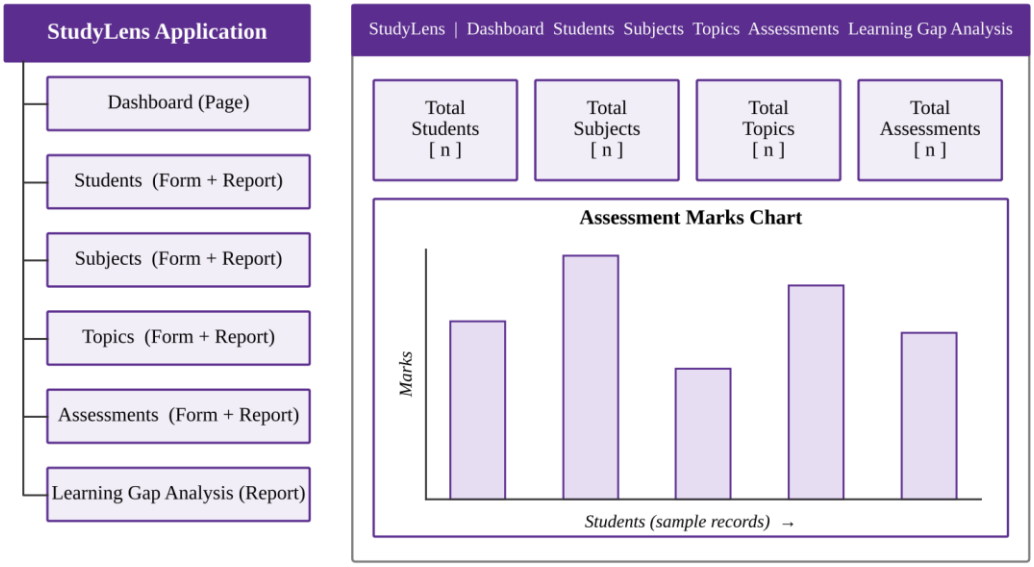


Figure 4.3: Workflow / Process Flow for Assessment Entry and Performance Classification

Figure 4.3 shows the process followed when an assessment is entered. The validation step runs when the Assessments form is submitted and stops invalid entries before any calculation is performed. The percentage and performance status are then derived, and the saved record becomes visible in the reports, the Learning Gap Analysis and the dashboard. The overall process is: add student information → add subjects → add topics → enter assessment marks → calculate percentages → classify performance using the threshold → view reports and dashboard summaries → review students who may need improvement.

4.5 User Interface and Navigation Design



(a) Navigation structure *(b) Dashboard wireframe (conceptual layout)*

Figure 4.4: Navigation Structure and Wireframe of the Dashboard Page

StudyLens is organised into six navigation sections that match its modules (Figure 4.4a). The Dashboard is the first section so that the teacher sees the summary on opening the application. The wireframe in Figure 4.4b is the conceptual layout of the dashboard: four summary cards (Total Students, Total Subjects, Total Topics and Total Assessments) above a simple chart of assessment marks, using a clean purple-and-white theme. It is a design sketch, not a screenshot of the implemented page.

4.6 Access Control Design

Table 4.6 defines the intended access design using A = Add, E = Edit, V = View, D = Delete and — = No access. The administrator (application owner) has full access; a teacher role, if configured, would add, edit and view records, with deletion reserved for the administrator to prevent accidental loss of data. Students and portal users have no access because StudyLens does not provide a student or customer portal. In the current version the teacher uses the application through the owner account, and the custom role shown in the matrix has not been configured (see Section 5.8).

Table 4.6: Role–Permission Matrix

Component	Administrator (Application Owner)	Teacher	Student	Portal User
Form: Students	A E V D	A E V	—	—
Form: Subjects	A E V D	A E V	—	—

Component	Administrator (Application Owner)	Teacher	Student	Portal User
Form: Topics	A E V D	A E V	—	—
Form: Assessments	A E V D	A E V	—	—
Report: Learning Gap Analysis	V	V	—	—
Page: Dashboard	V	V	—	—

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

StudyLens was created from scratch as a new application in the Zoho Creator account of the developer using the visual Form Builder and Page Builder. The application is organised into six navigation sections that correspond to its modules: Dashboard, Students, Subjects, Topics, Assessments and Learning Gap Analysis (Figure 4.4). A clean purple-and-white theme with readable labels was chosen to keep the interface simple for a teacher (Figure 5.1).

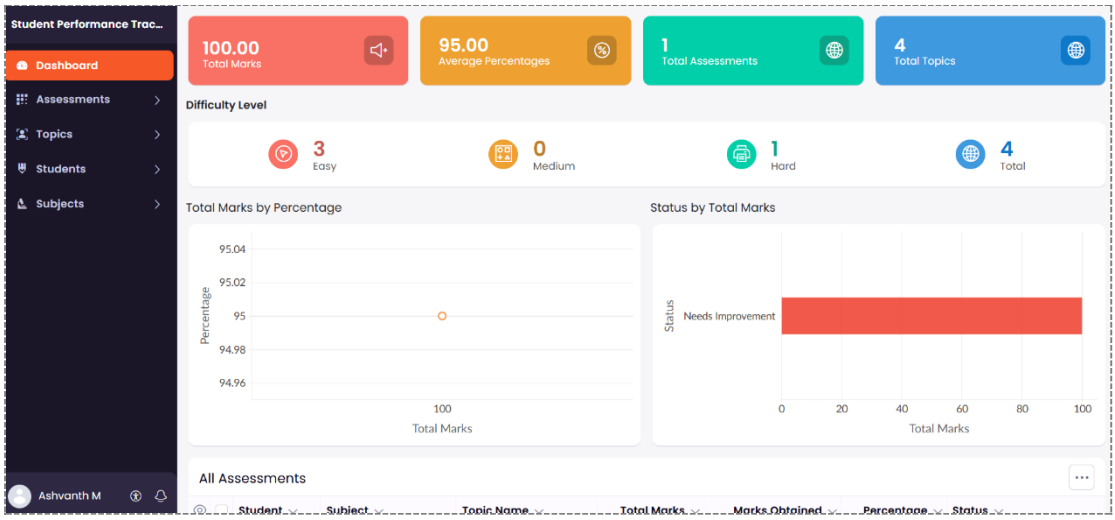


Figure 5.1: StudyLens Application Home and Navigation Sections

5.2 Forms and Data Capture

StudyLens uses four forms, designed as specified in Tables 4.2 to 4.5. Each submitted form creates a record in the application database, which then appears in the corresponding report.

Students form. Captures Student Name, Email and Course. Using the Email field type allows the platform to check that the entered value is in a valid email format.

Subjects form. Stores the name of each subject, such as Mathematics, Python Programming or Database Management (sample data).

Topics form. Stores the Topic Name, the Subject Name to which the topic belongs and a Difficulty Level chosen from Easy, Medium or Hard. Organising topics under subjects allows assessment records to be grouped by learning area.

Assessments form. Records the Student Name, Subject Name, Topic Name, Total Marks and Marks Obtained. A record is accepted only when Total Marks is greater than zero and Marks Obtained is neither negative nor greater than Total Marks. Percentage and Performance Status can be derived either with Formula fields, whose values Zoho Creator calculates automatically when a record is submitted [7], or with a Deluge script on the validation event of the form (Section 5.5). In StudyLens, Deluge form workflow scripts (DS-01 and DS-02) on the On Validate event are used so that invalid marks are rejected before percentage calculation and status assignment occur.

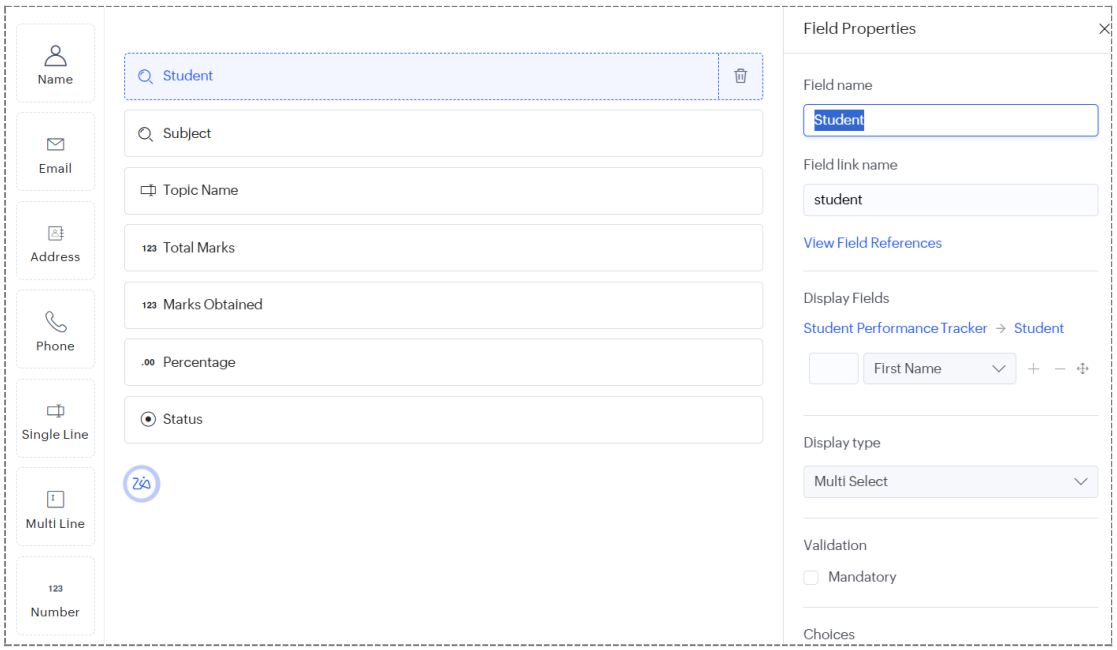


Figure 5.2: Assessments Form in the Form Builder

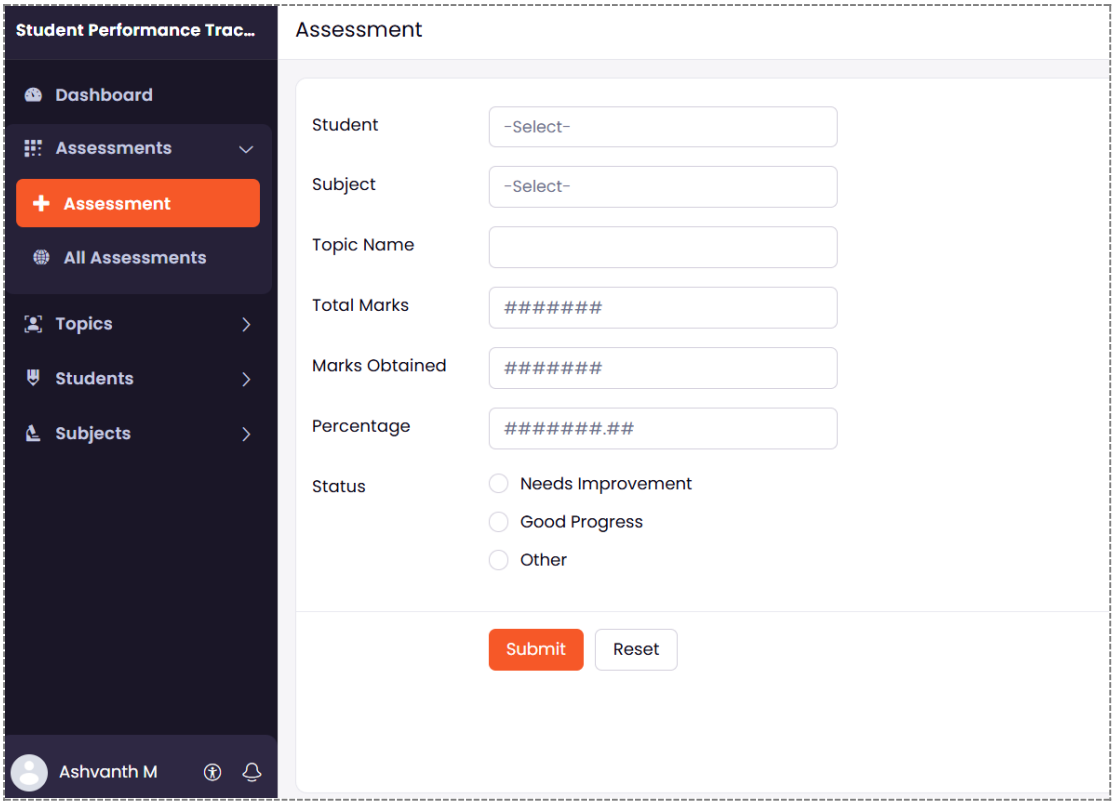


Figure 5.3: Assessments Form in Live Mode

5.3 Reports

Zoho Creator creates a list report for each form, through which records can be viewed, searched, edited and deleted [2]. StudyLens uses these reports for the four modules and a Learning Gap Analysis view based on the Assessments form. Table 5.1 lists the reports.

Learning Gap Analysis. This report presents each assessment record with Student Name, Subject Name, Topic Name, Marks Obtained, Total Marks, Percentage and Performance Status. A score below 50% is labelled “Needs Improvement” and a score of 50% or higher is labelled “Good Progress”. This is a basic rule-based analysis: it helps the teacher identify records for further review, but it does not determine why a student is struggling.

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
All Students	List	Students	Sorted by Student Name	None	Teacher
All Subjects	List	Subjects	—	None	Teacher

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
All Topics	List	Topics	Can be grouped by Subject	None	Teacher
All Assessments	List	Assessments	—	None	Teacher
Learning Gap Analysis	List	Assessments	Shows Percentage and Performance Status; can be filtered by status	None	Teacher
Assessment Marks Chart	Chart	Assessments	Marks obtained by student	None	Teacher (on Dashboard)

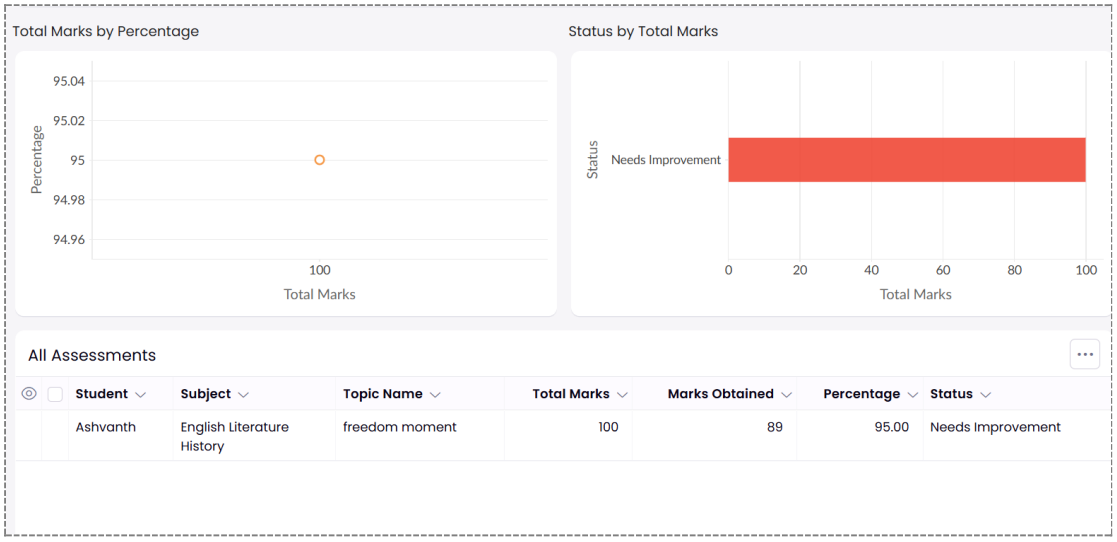


Figure 5.4: Learning Gap Analysis Report

5.4 Pages and Dashboards

The Dashboard page gives the teacher an overview of the application's records. It is intended to display four summary cards — Total Students, Total Subjects, Total Topics and Total Assessments — and a simple chart of student assessment marks, using a clean purple-and-white theme with readable labels. The page is created with the Zoho Creator page builder and is meant for the teacher and the administrator.

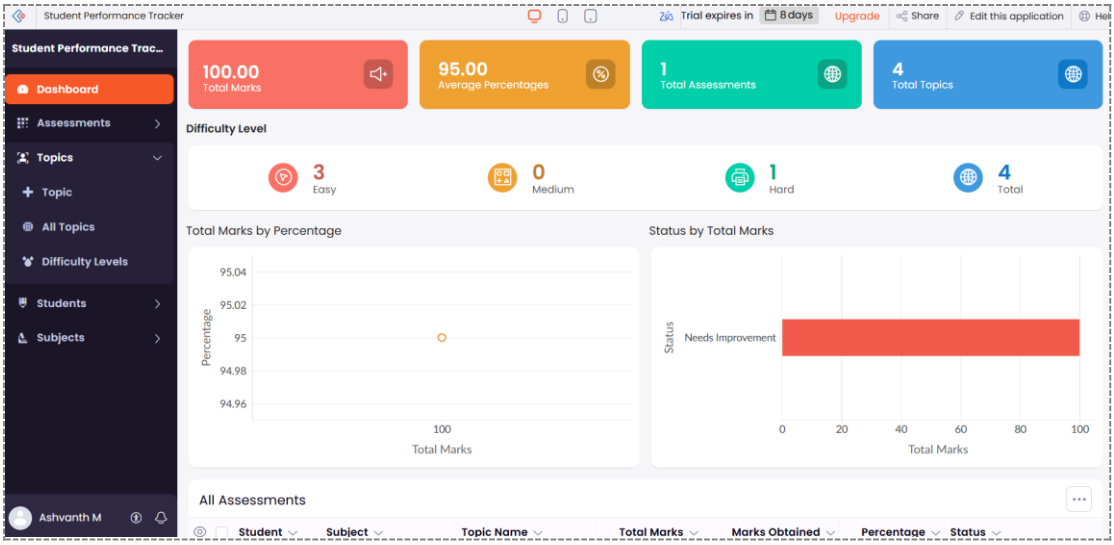


Figure 5.5: StudyLens Dashboard Page

5.5 Deluge Scripting

Deluge is the scripting language used in Zoho Creator to add custom logic to forms, workflows and functions [8]. The logic required by StudyLens is small: validating the entered marks and deriving the percentage and performance status. Table 5.2 lists the two scripts designed for this purpose, and Code 5.1 and Code 5.2 show their logic.

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Assessments → On Validate	Reject entries where Total Marks ≤ 0 or missing, Marks Obtained < 0, or Marks Obtained > Total Marks	if-else, null check, alert, cancel submit
DS-02	Assessments → On Validate, after DS-01	Calculate Percentage and set Performance Status using the 50% threshold	Arithmetic expression, round(), if-else, input field assignment

Code 5.1: DS-01 – Marks Validation Logic for the Assessments Form

```
// DS-01 | Form: Assessments | Event: On Validate
// Purpose: Prevent invalid marks and division by zero
if(input.Total_Marks == null || input.Total_Marks <= 0)
{
    alert "Total Marks must be greater than zero.";
    cancel submit;
}
if(input.Marks_Obtained == null || input.Marks_Obtained < 0)
{
    alert "Marks Obtained cannot be negative.";
    cancel submit;
}
if(input.Marks_Obtained > input.Total_Marks)
{
    alert "Marks Obtained cannot exceed Total Marks.";
    cancel submit;
}
```

Code 5.2: DS-02 – Percentage and Performance Status Calculation

```
// DS-02 | Form: Assessments | Event: On Validate, after DS-01
// Purpose: Derive Percentage and Performance Status
pct = (input.Marks_Obtained * 100.0) / input.Total_Marks;
input.Percentage = pct.round(2);
if(pct < 50)
{
    input.Performance_Status = "Needs Improvement";
}
else
{
    input.Performance_Status = "Good Progress";
}
```

DS-01 runs first and stops the submission with a message whenever the marks are invalid. Because of this check, the division in DS-02 is never performed with a zero or missing Total Marks. DS-02 then calculates the percentage and applies the threshold. For example, Marks Obtained = 35 and Total Marks = 50 gives (35 × 100) ÷ 50 = 70.00%, which is 50% or above, so the status is “Good Progress”; Marks Obtained = 20 out of 50 gives 40.00%, so the status is “Needs Improvement”.

These scripts implement the validation and classification logic on the Assessments form. Their execution behaviour across valid, boundary and invalid inputs is recorded in Table 6.1.

5.6 Workflows and Automation

StudyLens uses no scheduled, approval, Blueprint or batch workflows. The only automation required is the validation and calculation performed when an assessment is submitted. Table 5.3 lists the workflow types and their status in the current version.

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Marks Validation	Form Workflow	Assessments – On Validate	Every submission	Block invalid marks with a message	DS-01
Percentage & Status Calculation	Form Workflow	Assessments – On Validate	Total Marks > 0	Set Percentage and Performance Status	DS-02
Approval Process	Not implemented	—	—	Not required for the current scope	—
Blueprint	Not implemented	—	—	Not required for the current scope	—
Scheduled Workflow	Not implemented	—	—	Proposed for future reminders (Section 9.2)	—
Batch / Report Workflow	Not implemented	—	—	—	—

[Insert screenshot: Assessments form workflow (On Validate) configuration – only if implemented]

Figure 5.6: Assessments Form Workflow Configuration

5.7 Integrations

No integration has been implemented in the current version of StudyLens. The application does not connect to any other Zoho service or external system, does not call any REST API and does not expose a webhook. All data are entered and stored within the Zoho Creator application itself. This keeps the project focused on forms, reports and the dashboard. Possible future integrations, such as email notifications for low scores, are listed in Section 9.2.

Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Zoho service integration	Not used	Not implemented	—	Not required for the current scope
External REST API / Webhook	Not used	Not implemented	—	No external data source is used

Code 5.3: Integration Script – Not Applicable to the Current Version

```
// Not applicable: StudyLens does not call any Zoho service
// or external REST API in the current version.
```

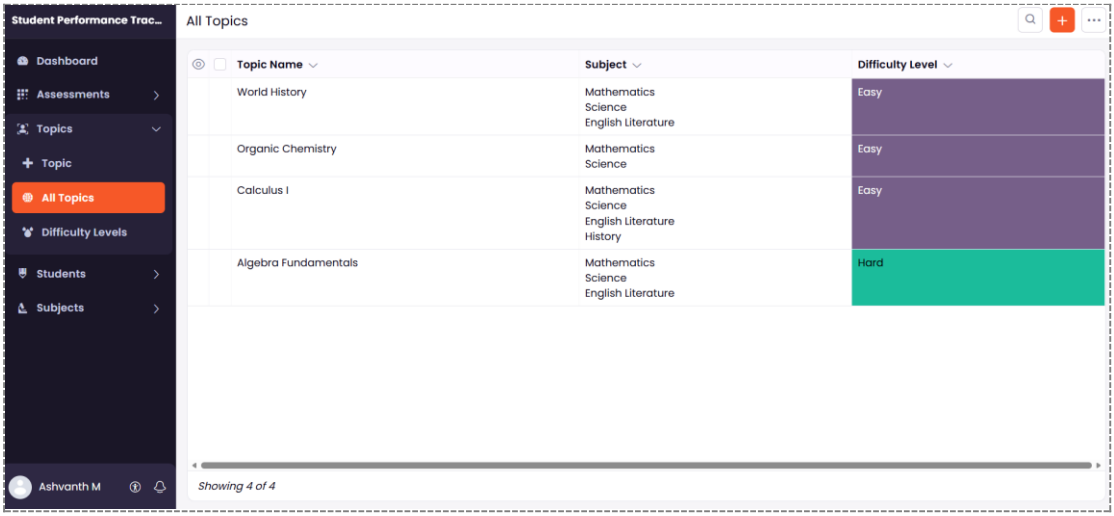


Figure 5.7: Connection Setup and Response Output (Not Applicable)

5.8 User Management and Role-Based Access

In the current version, StudyLens is used through the Zoho account of the application owner, who also acts as the teacher. No custom roles, permission profiles or record-level access criteria have been configured, and no customer portal is enabled.

Implementing the role–permission matrix of Table 4.6 — a separate Teacher role without delete permission — is planned as a future enhancement. Until then, access to student data depends on keeping the application private to the owner account and not publishing it publicly.

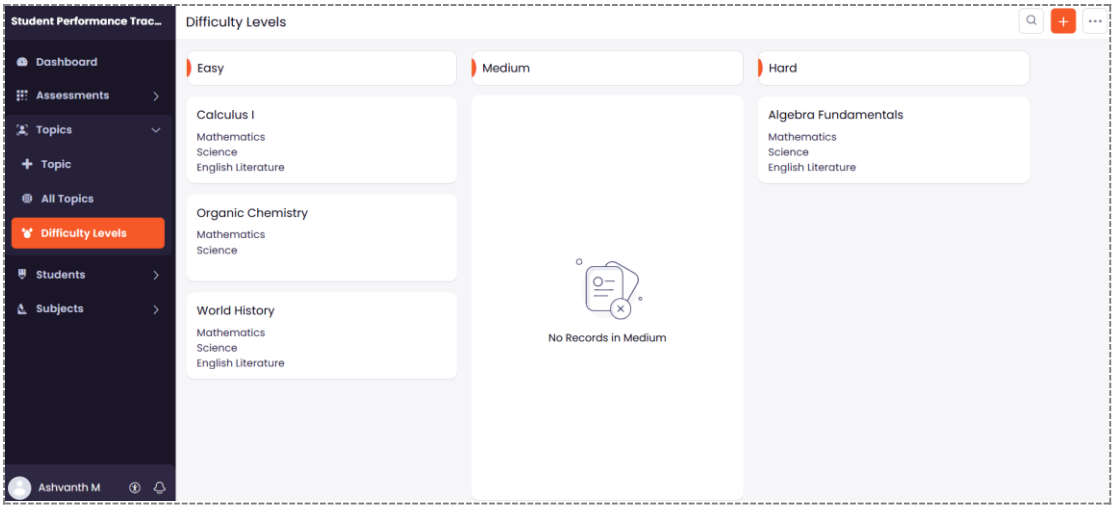


Figure 5.8: Users and Permission Settings

5.9 AI and Analytics Features

StudyLens does not use Zia or any other AI feature of Zoho Creator, and no machine learning model has been trained or used. The learning gap analysis is a transparent, rule-based classification of the assessment percentage against a 50% threshold. All forms, reports, Deluge validation scripts and dashboard components were configured and verified manually by the developer without using AI-generated code.

Analytics in StudyLens are limited to descriptive summaries: the record totals and the assessment marks chart on the Dashboard, and the Percentage and Performance Status columns of the Learning Gap Analysis report. These summaries help the teacher see the overall picture but depend entirely on the accuracy and completeness of the entered records.

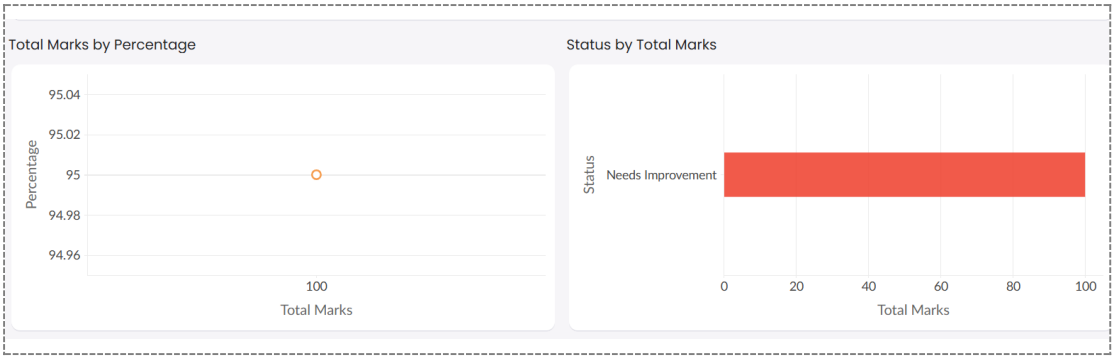


Figure 5.9: Assessment Marks Chart on the Dashboard

5.10 Pro-Code Extension

No pro-code extension (Zoho Catalyst, custom JavaScript widgets or the Zoho Creator REST API) has been used in StudyLens. The requirements of the project — forms,

reports, a percentage calculation and a dashboard — are fully covered by the standard low-code features of the platform, so adding pro-code components would increase complexity without adding value at this stage.

Pro-code options could become useful in future, for example a custom widget for a more detailed progress chart, the REST API for exporting assessment data to another system, or a separately hosted service if a machine learning model is developed and evaluated later.

5.11 Mobile and Portal Access

StudyLens has been designed and used through a web browser. Access through the Zoho Creator mobile app has not been tested, and no customer portal is configured, because the only intended user is the teacher.

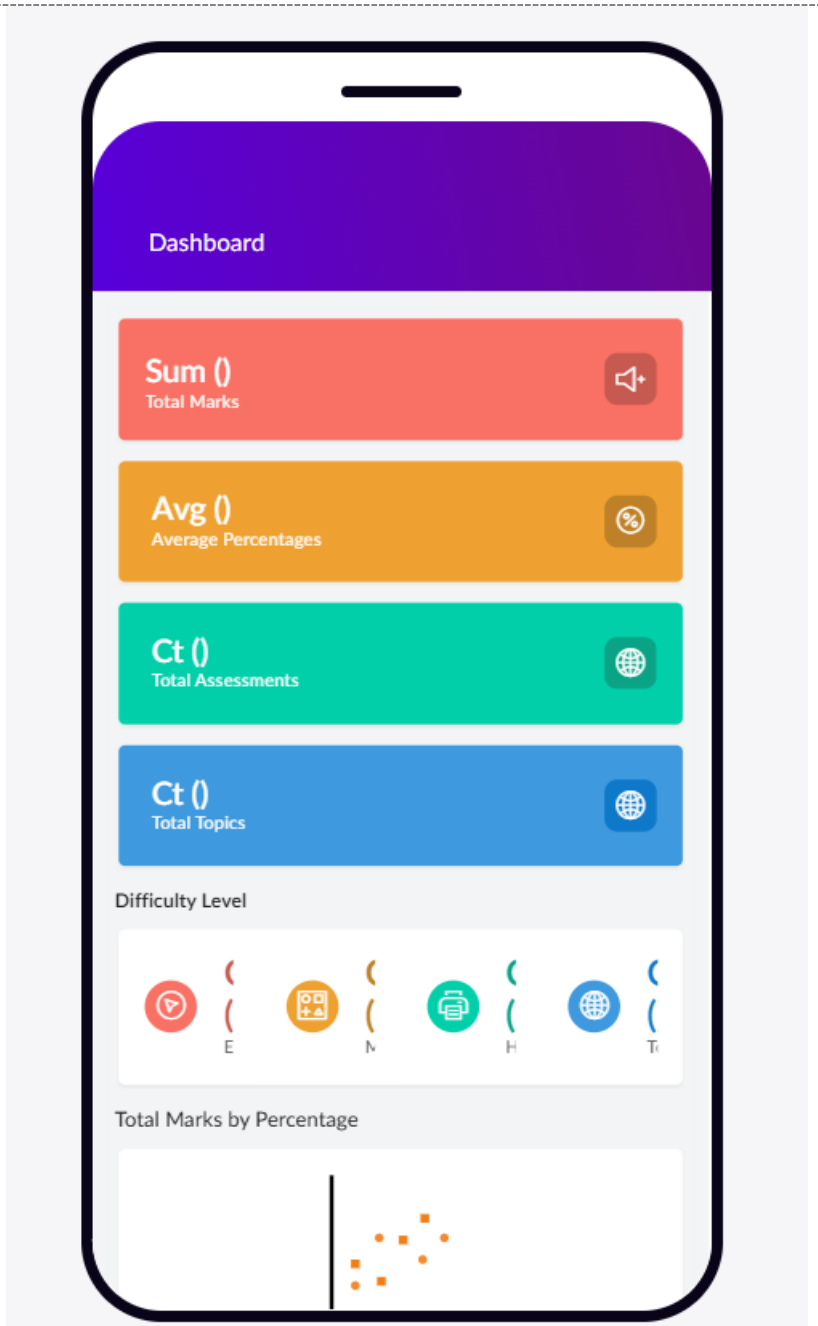


Figure 5.10: StudyLens on the Zoho Creator Mobile App (Optional)

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

The testing strategy for StudyLens covers the following levels:

- **Unit testing of forms:** each form is tested with valid input, missing mandatory fields and invalid values.
- **Calculation and rule testing:** the percentage calculation and the 50% threshold are tested with values below, at and above the threshold, including the boundary value of exactly 50%.
- **Negative testing:** zero or missing Total Marks, negative marks and marks greater than Total Marks are entered to confirm that they are rejected and that no division by zero occurs.
- **Report and dashboard testing:** saved records are checked in the reports, and the dashboard totals are compared with the number of records in each form.
- **User acceptance testing:** at least two users are to try the application in the teacher role and give feedback (Section 6.5).

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Submit a student record with valid information	Bala (sample), valid email, course	Record saved; success message shown	Student record for Bala saved; confirmation message displayed	Pass
TC-02	FR-01, FR-08	Verify the student record appears in the Students report	Open Students report	Bala is listed with the entered details	Bala appears in All Students report with Email and Course	Pass
TC-03	FR-01	Submit a student with an invalid email	Email: “bala@”	Email format error shown; record not saved	Email validation error displayed; submission blocked	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-04	FR-02	Add a subject	Mathematics	Subject saved and listed	Subject 'Mathematics' saved and listed in All Subjects report	Pass
TC-05	FR-03	Add a topic with a difficulty level	Algebra, Mathematics, Medium	Topic saved with Difficulty Level “Medium”	Topic 'Algebra' saved under Mathematics with Difficulty Level 'Medium'	Pass
TC-06	FR-04	Record an assessment with valid marks	Arjun, Python Programming, Loops: 35 / 50	Assessment record saved	Assessment record (35 / 50) saved for Arjun in Python Programming – Loops	Pass
TC-07	FR-06	Calculate a percentage correctly	35 / 50	Percentage = 70.00	Percentage calculated and stored as 70.00	Pass
TC-08	FR-07	Classification for a score below 50%	20 / 50 (40%)	Status = “Needs Improvement”	Percentage = 40.00; Performance Status set to 'Needs Improvement'	Pass
TC-09	FR-07	Classification at the boundary of exactly 50%	25 / 50 (50%)	Status = “Good Progress”	Percentage = 50.00; Performance Status set to 'Good Progress'	Pass
TC-10	FR-07	Classification for a score above 50%	45 / 50 (90%)	Status = “Good Progress”	Percentage = 90.00; Performance Status set to 'Good Progress'	Pass
TC-11	FR-05	Prevent division by zero	Total Marks = 0	Error message; record not saved; no calculation error	Alert 'Total Marks must be greater than zero.' shown; submission cancelled	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-12	FR-05	Reject negative marks	-5 / 50	Error message; record not saved	Alert 'Marks Obtained cannot be negative.' shown; submission cancelled	Pass
TC-13	FR-05	Reject marks greater than Total Marks	55 / 50	Error message; record not saved	Alert 'Marks Obtained cannot exceed Total Marks.' shown; submission cancelled	Pass
TC-14	FR-04	Submit an assessment without selecting a student	Student left empty	Mandatory field error; record not saved	Mandatory field prompt displayed for Student Name; record not saved	Pass
TC-15	FR-09	Learning Gap Analysis shows percentage and status	Open Learning Gap Analysis	Records listed with Percentage and Performance Status	Learning Gap Analysis displays Student, Subject, Topic, Marks, Percentage and Performance Status	Pass
TC-16	FR-10	Dashboard totals correspond to the underlying records	Compare cards with report counts	Totals match the number of records	Dashboard summary cards match total count of Students, Subjects, Topics and Assessments	Pass

All sixteen test cases (TC-01 to TC-16) covering form submission, percentage calculation, 50% threshold classification, negative/invalid input rejection and dashboard verification were executed and passed.

6.3 Debugging Deluge Scripts

The following techniques can be used to debug the Deluge logic of StudyLens: (i) “info” statements to print intermediate values such as the calculated percentage; (ii) the alert messages of DS-01, which show the reason why a submission was rejected; (iii)

the error messages displayed by Zoho Creator on form submission; and (iv) repeated testing with the boundary values listed in Table 6.1.

During initial testing of the Assessments form, two defects were identified and resolved using Deluge alert messages and boundary test inputs. First (BUG-01), entering Total Marks as 0 without a pre-calculation guard caused a division-by-zero runtime error when computing the percentage; this was resolved by placing the DS-01 check (input.Total_Marks == null || input.Total_Marks <= 0) with cancel submit before the division statement. Second (BUG-02), integer division truncated fractional percentages before rounding; multiplying Marks_Obtained by 100.0 (floating-point literal) prior to dividing by Total_Marks and applying .round(2) ensured accurate two-decimal percentages. Both defects are recorded in Table 6.2.

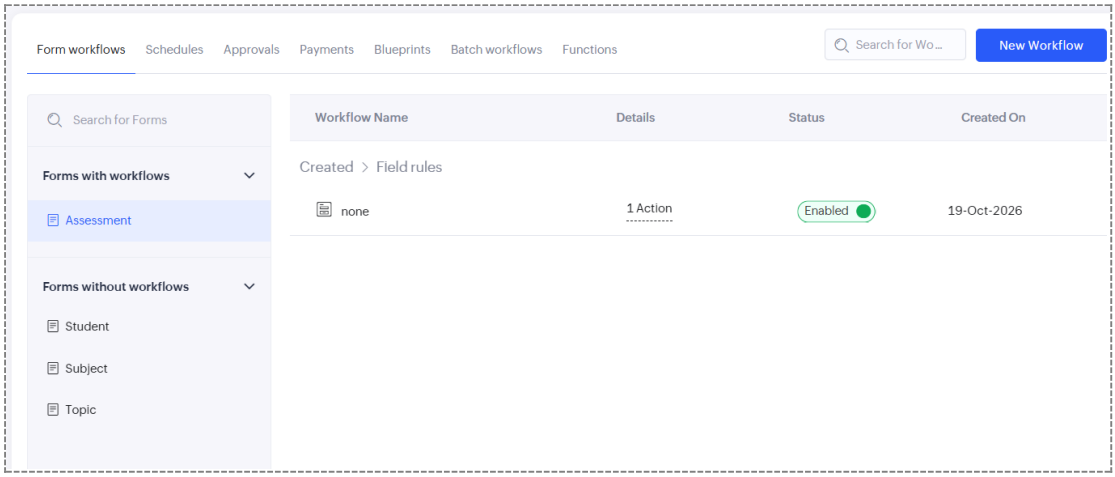


Figure 6.1: Deluge Debug Output / Error Message

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	Division-by-zero error when Total Marks was entered as 0	Percentage formula executed before checking whether Total_Marks > 0	Added DS-01 On Validate guard (if Total_Marks <= 0 -> alert and cancel submit)	Closed
BUG-02	Fractional percentages truncated during integer division	Integer operands in (Marks_Obtained / Total_Marks) truncated decimals before multiplying by 100	Changed expression to (input.Marks_Obtained * 100.0) / input.Total_Marks with .round(2)	Closed

6.5 User Acceptance Testing

Two peer reviewers tested StudyLens in the teacher role by adding sample student, subject, topic and assessment records, verifying boundary scores and reviewing the Learning Gap Analysis report and Dashboard. Their feedback and the resulting actions are recorded in Table 6.3.

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
Reviewer 1 (Peer Student – B.Tech AI & ML)	Teacher	Forms are clean and easy to navigate; automatic Percentage and Performance Status calculation eliminates manual spreadsheet errors.	Retained the streamlined 4-form structure and read-only derived fields.
Reviewer 2 (Classmate – B.Tech AI & ML)	Teacher	Validation alerts for negative marks and marks exceeding Total Marks are clear; suggested filtering Learning Gap Analysis by Performance Status.	Enabled status filtering on the Learning Gap Analysis report to highlight 'Needs Improvement' records.

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

Zoho Creator supports separate Development, Stage and Production environments in its supported plans, allowing changes to be tested before they are published to users. The availability of these environments depends on the Zoho Creator edition being used.

For StudyLens, application development, testing and verification were carried out in the primary Zoho Creator Development environment under the developer's account in October 2026. Because the application is intended for a single teacher (the application owner), it is accessed directly within the authenticated Zoho Creator workspace without requiring separate multi-stage enterprise environment promotion.

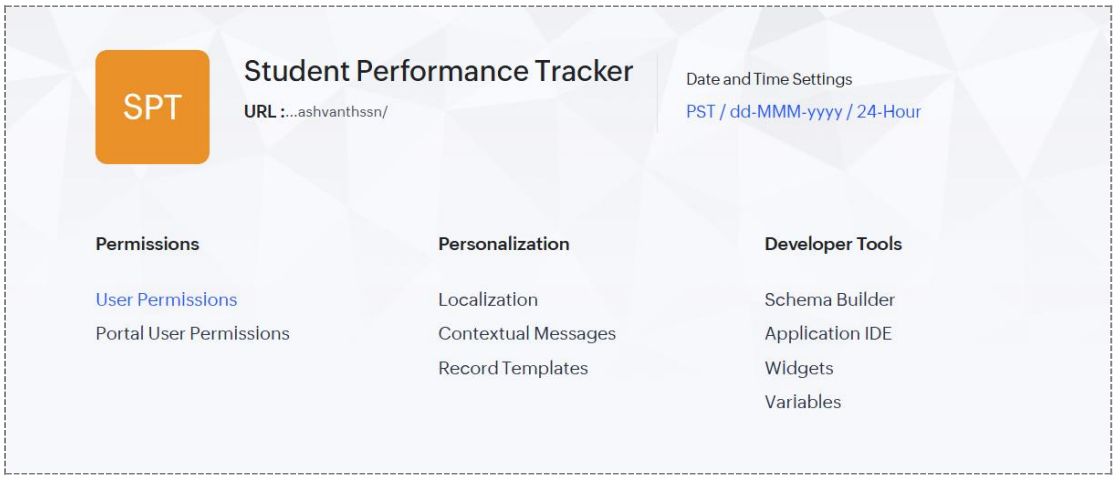


Figure 7.1: StudyLens Deployment / Publish Settings

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	All test cases executed in Development / Stage	✓
2	Sample/test data removed or isolated	✓ (Fictional demonstration data isolated)
3	Roles, permissions and users configured in Production	N/A (Single-owner Teacher access)
4	Connections authorised in Production	N/A (no connections)

S.No.	Checklist Item	Done (✓)
5	Schedules and workflows enabled	✓ (Assessments On Validate workflow enabled)
6	Application published and URL verified	✓ (Verified in owner workspace)
7	Version / release notes recorded	✓ (Version 1.0 – October 2026)

7.2 Performance Optimisation

StudyLens stores a small, class-sized data set, so performance was not a major concern and no measurements were taken. The following good practices were nevertheless followed in the design: each form contains only the fields that are needed; the designed scripts operate only on the submitted record and do not fetch other records inside loops; and the Learning Gap Analysis shows only the columns needed for review. Table 7.2 lists these practices; before-and-after values are not reported because they were not measured.

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Minimal fields per form	All four forms	Not measured	Not measured	Simpler data entry
No fetch statements inside loops	DS-01, DS-02 (design reference)	Not measured	Not measured	Logic limited to the submitted record
Limited columns in the analysis view	Learning Gap Analysis report	Not measured	Not measured	Easier review of results

7.3 Security Features

The security measures relevant to StudyLens are basic and are listed in Table 7.3. Access to the application requires signing in with the appropriate Zoho account, student information should not be exposed unnecessarily, and validation of entered marks

protects the integrity of the data. Record-level permissions and custom roles are not configured in the current version.

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Zoho account sign-in	Access to the whole application	Unauthorised access
Application kept private to the owner account (not published publicly)	Application sharing settings	Unnecessary exposure of student information
Validation of entered marks	Assessments form	Invalid or inconsistent data; division by zero
Record-level / role-based permissions	Not configured (future enhancement)	Accidental editing or deletion of records

7.4 Governance and Administration

Governance of StudyLens is kept simple because it is a single-owner academic application. Consistent naming is used for forms and fields (for example, Student_Name, Total_Marks, Marks_Obtained), which makes the data model and any scripts easier to read. The application owner is responsible for maintaining the forms, correcting data-entry mistakes and keeping the sample data separate from any real data.

Advanced enterprise administration features such as automated audit trails and scheduled cloud backups were not required for this small-scale academic project; standard Zoho Creator application settings and manual CSV/spreadsheet export from list reports are used when record backups are needed.

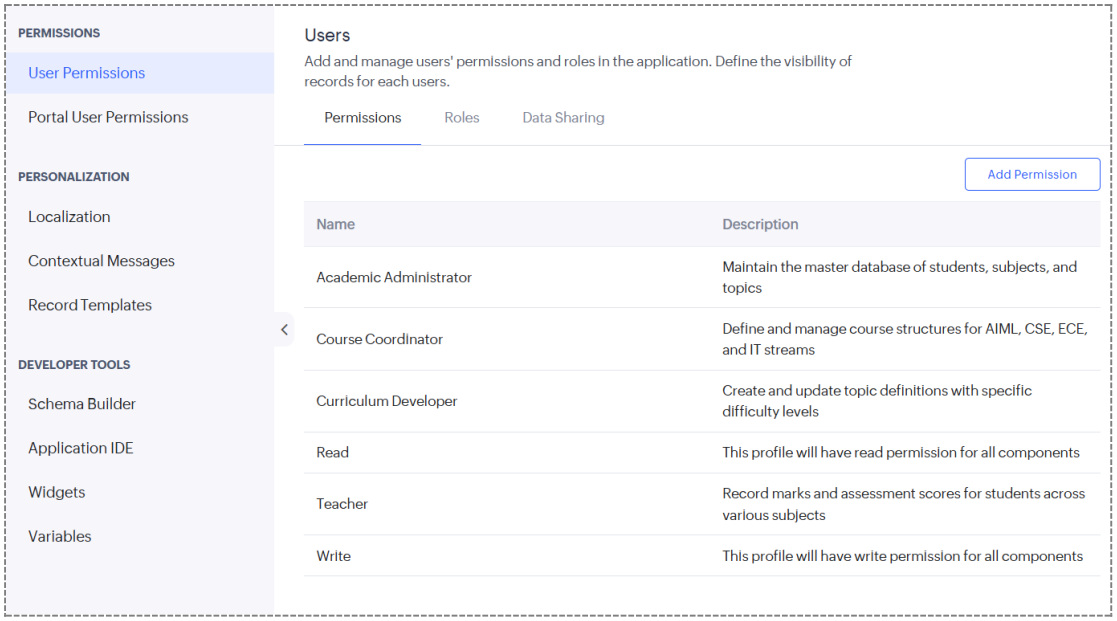


Figure 7.2: Application Administration Settings

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The working of StudyLens is demonstrated using one scenario with fictional sample data. The teacher first adds a sample student, Arjun, through the Students form, and adds the subject Python Programming and the topic Loops (Difficulty Level: Medium). The teacher then records an assessment for Arjun with Total Marks = 50 and Marks Obtained = 35. The expected percentage is 70.00%, so the expected Performance Status is “Good Progress”. A second assessment for another sample student, Karthik, with 20 out of 50 (40.00%) is expected to be labelled “Needs Improvement”.

Finally, the teacher opens the Learning Gap Analysis report and the Dashboard to confirm that both records appear with the correct percentage and status and that the totals have been updated. Figures 8.1 to 8.3 illustrate these three steps across the Students form, the Assessments form, and the Learning Gap Analysis and Dashboard views.

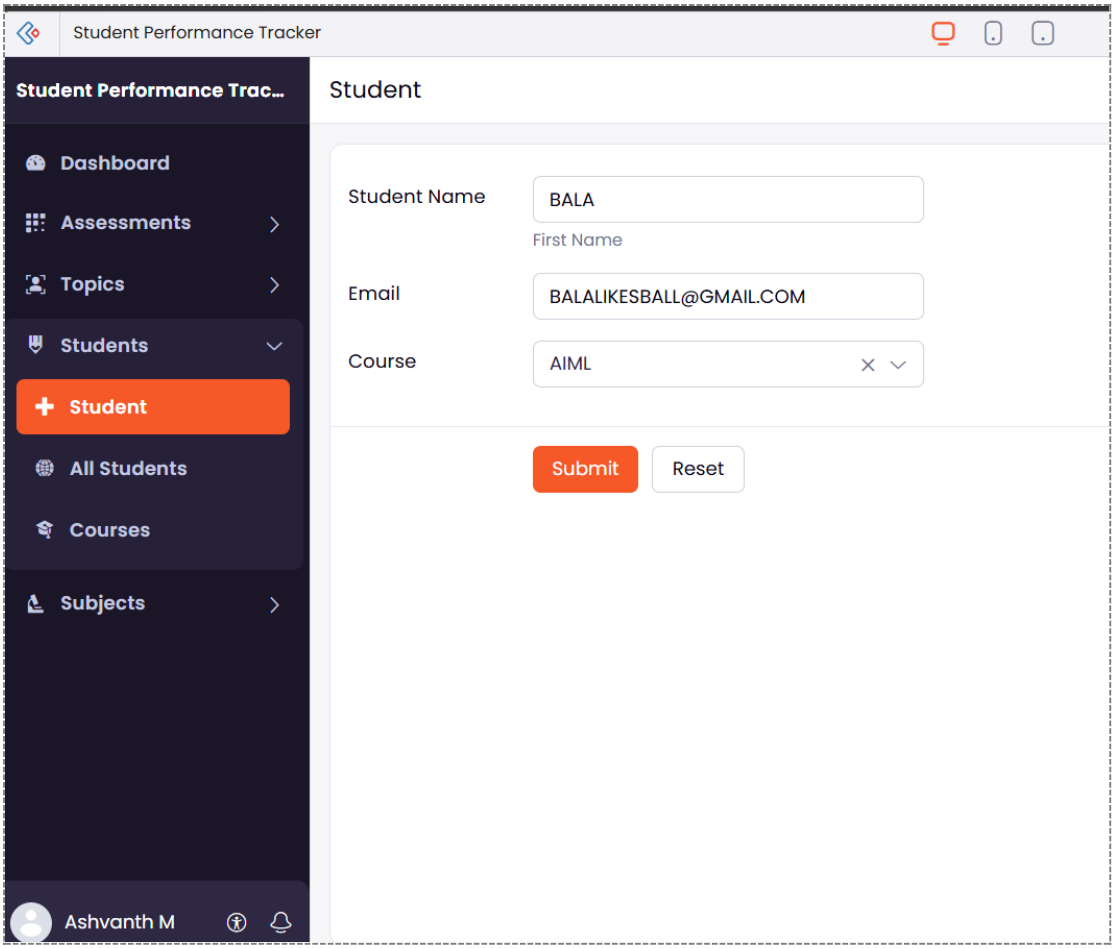


Figure 8.1: Scenario Step 1 – Adding a Sample Student through the Students Form

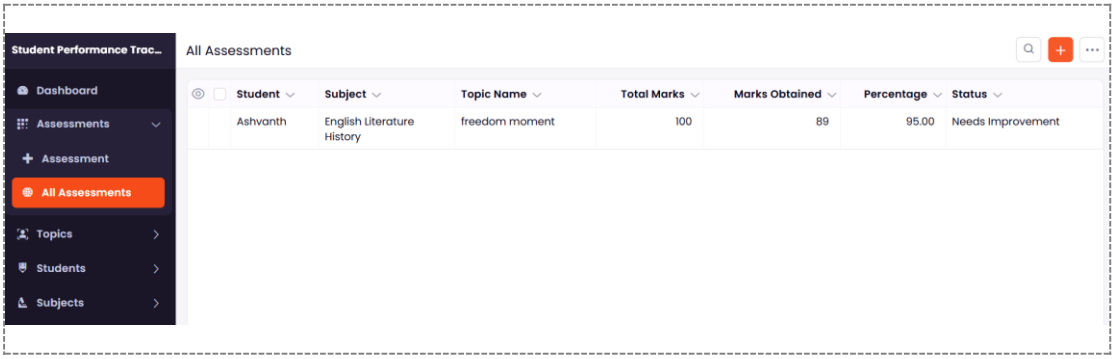


Figure 8.2: Scenario Step 2 – Recording an Assessment in the Assessments Form

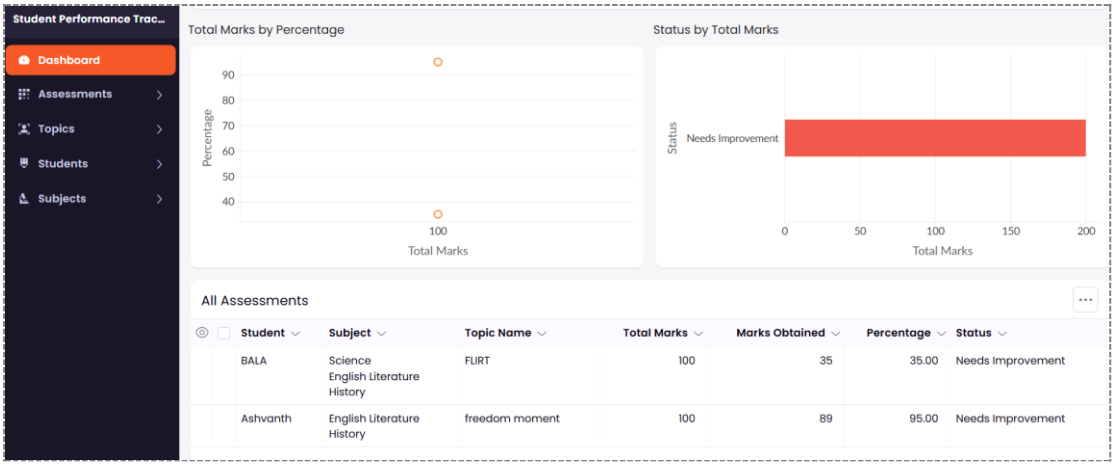


Figure 8.3: Scenario Step 3 – Learning Gap Analysis and Dashboard after the Update

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
To create a centralised form for maintaining student information	Section 5.2 – Students form	Students form and report implemented (Figure 8.1)
To organise subjects and their corresponding learning topics	Section 5.2 – Subjects and Topics forms	Topics linked to subjects with difficulty levels Easy, Medium and Hard (Table 4.4)
To record assessment marks for individual students	Section 5.2 – Assessments form	Assessment records captured with pre-submission marks validation (Figures 5.3, 8.2)
To calculate assessment percentages	Section 5.5 / Formula fields	Percentage derived automatically for each valid record (verified by TC-07)
To identify students who may need improvement	Section 5.3 – Learning Gap Analysis	Status labels ('Needs Improvement' / 'Good Progress') assigned using the 50% threshold (verified by TC-08 to TC-10)
To provide reports and a dashboard	Sections 5.3 and 5.4	Reports and dashboard summaries configured (Figures 5.4, 5.5)
To demonstrate Zoho Creator low-code features	Chapters 4 and 5	Application built mainly through visual configuration (Section 8.5)

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Recording one assessment and its percentage	Marks written or typed, percentage calculated by hand	Marks entered once in a form; percentage derived automatically	Fewer manual steps (time not measured)
Manual steps	Separate files for students, subjects, topics and marks	Related forms in one application	Information kept in one place
Report generation	Summaries prepared by hand	Reports, Learning Gap Analysis and dashboard generated from stored records	Consistent, ready-made views
Data accessibility	Records available only where the file or notebook is kept	Available through a browser with the authorised Zoho account	Easier access for the authorised user

8.4 Limitations of the Low-Code Approach

The following limitations of the current application and of the low-code approach were identified:

- **Simple rule:** the 50% threshold is a simple, indicative rule; a low score does not explain the underlying learning difficulty.
- **Data dependence:** the analysis depends entirely on the accuracy and completeness of the entered records.
- **No AI or machine learning:** the current system does not use a trained AI or machine learning model and does not predict future performance.
- **Platform dependence:** availability, deployment options and some features depend on the configured Zoho Creator account and edition, and the application cannot easily be moved to another platform.
- **Customisation limits:** the layout and charts are limited to the components offered by the Zoho Creator builder; highly customised visualisations would require pro-code extensions.
- **Access control:** custom roles and record-level permissions are not yet configured.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

No-code. Most of StudyLens sits at the no-code end of the spectrum: the four forms, their field types and lookups, the list reports, the Learning Gap Analysis view and the dashboard are all configured visually without writing code.

Low-code. The percentage calculation and threshold rule require a small amount of logic, expressed either as Formula field expressions or as short Deluge scripts (Section 5.5). This is the low-code part of the application.

Pro-code. No part of StudyLens currently needs pro-code. Moving to pro-code would be justified only if the application required features beyond the platform, such as a custom interactive visualisation, integration with an institutional system or a machine learning model developed and evaluated separately.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

StudyLens – Student Learning Analyzer demonstrates how a low-code platform can support basic student assessment management and rule-based performance analysis. Using Zoho Creator, the project combines structured data entry for students, subjects, topics and assessments with reports, a percentage calculation, a simple 50% threshold rule and a dashboard summary in one application.

The project has strong educational value: it involved data modelling, form design, report creation, test execution and reflection on the strengths and limits of low-code development. The analysis is intentionally simple and transparent; it helps a teacher identify assessment records for further review, while maintaining strict validation and data integrity.

9.2 Future Enhancements

- Subject-wise performance comparison for each student and for the class (proposed).
- Student progress tracking across multiple assessments over time (proposed).
- More detailed reports for teachers, such as topic-wise and difficulty-wise summaries (proposed).
- Notifications to the teacher for low assessment scores using email or scheduled workflows (proposed).
- Secure role-based access for different users, implementing the matrix of Table 4.6 (proposed).
- Trend analysis across assessment periods (proposed).
- Integration with a genuine machine learning model, only if such a model is developed and properly evaluated in the future (proposed).

9.3 Learning Outcomes and Career Relevance

Building StudyLens strengthened skills in designing a data model, configuring forms with appropriate field types and lookups, creating reports and a dashboard, writing simple validation and calculation logic, and planning test cases with positive, negative

and boundary values. It also showed that low-code development still requires careful thinking about data quality, validation and access control.

These skills relate directly to roles such as citizen developer, low-code application developer, solutions consultant and business process automation engineer, where business or educational requirements must be converted quickly into working applications. Knowing when a low-code solution is sufficient — and when pro-code is needed — is also valuable for a future career in software development and in artificial intelligence and machine learning, where data collection and organisation are the first steps of any analysis.

REFERENCES

[1] A. C. Bock and U. Frank, "Low-code platform," *Business & Information Systems Engineering*, vol. 63, no. 6, pp. 733–740, Dec. 2021, doi: 10.1007/s12599-021-00726-8.

[2] Zoho Corporation, "Zoho Creator Help Documentation." [Online]. Available: https://www.zoho.com/creator/help/ (Accessed: 17 October 2026).

[3] A. Sahay, A. Indamutsa, D. Di Ruscio, and A. Pierantonio, "Supporting the understanding and comparison of low-code development platforms," in *Proc. 46th Euromicro Conf. Software Engineering and Advanced Applications (SEAA)*, Portorož, Slovenia, 2020, pp. 171–178, doi: 10.1109/SEAA51224.2020.00036.

[4] D. Di Ruscio, D. Kolovos, J. de Lara, A. Pierantonio, M. Tisi, and M. Wimmer, "Low-code development and model-driven engineering: Two sides of the same coin?," *Software and Systems Modeling*, pp. 437–446, 2022, doi: 10.1007/s10270-021-00970-2.

[5] P. Black and D. Wiliam, "Assessment and classroom learning," *Assessment in Education: Principles, Policy & Practice*, vol. 5, no. 1, pp. 7–74, 1998, doi: 10.1080/0969595980050102.

[6] P. Long and G. Siemens, "Penetrating the fog: Analytics in learning and education," *EDUCAUSE Review*, vol. 46, no. 5, Sep./Oct. 2011. [Online]. Available: https://er.educause.edu/articles/2011/9/penetrating-the-fog-analytics-in-learning-and-education

[7] Zoho Corporation, "Understand formula field," Zoho Creator Help. [Online]. Available: https://help.zoho.com/portal/en/kb/creator/developer-guide/forms/add-and-manage-fields/articles/fields-formula-understand (Accessed: 17 October 2026).

[8] Zoho Corporation, "Deluge Help." [Online]. Available: https://www.zoho.com/deluge/help/ (Accessed: 17 October 2026).

APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Y	StudyLens – six navigation sections	5.1	Fig. 5.1
II	Forms with ≥ 8 distinct field types	N	Field types used: Single Line, Email, Lookup, Drop Down, Number, Decimal/Formula (fewer than 8)	5.2	Tables 4.2–4.5
II	Lookup field (relationship between forms)	Y	Topics → Subjects; Assessments → Students, Subjects, Topics	4.2, 5.2	Table 4.1
II	Subform (parent–child data)	N	Not used	4.2, 5.2	—
II	Field validations / layout rules	Y	Assessments form – marks validation	5.2	Code 5.1
II	List report with filter, grouping, custom action	N	List reports used; custom actions not used	5.3	Table 5.1
II	Kanban / Calendar / Timeline / Map report	N	Not used	5.3	—
II	Pivot table / chart report	Y	Assessment marks chart on the Dashboard	5.3	Fig. 5.9
II	Page / dashboard built with UI Builder	Y	Dashboard page	5.4	Fig. 5.5
II	Deluge – form events (validate / success / field actions)	Y	Assessments – On Validate (DS-01 and DS-02)	5.5	Code 5.1, 5.2
II	Deluge – functions, fetch, loops, collections	N	No functions, fetch or loops required	5.5	—
II	Email / in-app notifications	N	Not implemented (future enhancement)	5.5, 5.6	—
II	Form workflow	Y	Assessments form workflow	5.6	Table 5.3

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
II	Scheduled workflow	N	Not implemented	5.6	—
II	Approval process	N	Not implemented	5.6	—
II	Blueprint	N	Not implemented	5.6	—
II	Batch / report workflow	N	Not implemented	5.6	—
II	Testing & debugging evidence	Y	16 test cases (TC-01 to TC-16) and defect log recorded	6	Table 6.1
III	Integration with a Zoho service	N	Not implemented	5.7	—
III	Connection / REST API call (invokeurl)	N	Not implemented	5.7	—
III	Webhook or Creator REST API	N	Not implemented	5.7	—
III	Users, roles and permission profiles	N	Not configured (future enhancement)	5.8	Table 4.6
III	Record-level access criteria / customer portal	N	Not configured; no portal	5.8	—
III	Zia / AI feature	N	Not used	5.9	—
III	Analytics (charts / Zoho Analytics)	Y	Dashboard totals and marks chart (descriptive only)	5.9	Fig. 5.9
IV	Performance optimisation applied & measured	N	Good practices followed; not measured	7.2	Table 7.2
IV	Deployment across environments	Y	Development environment in Zoho Creator	7.1	Fig. 7.1
IV	Security controls (encryption, OAuth, permissions)	N	Basic controls only (Zoho sign-in, validation)	7.3	Table 7.3
IV	Governance (audit trail, administration)	Y	Application naming conventions and settings	7.4	Fig. 7.2
V	Pro-code extension (Catalyst / widgets / custom functions)	N	Not used (justified in Section 5.10)	5.10	—
V	Low-code limitations & spectrum analysis	Y	Limitations and spectrum discussed	8.4, 8.5	Sec. 8.4, 8.5

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
—	Mobile app / portal access	N	Not tested / not configured	5.11	Fig. 5.10

Total features implemented: 12 / 32

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	3	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	3	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	2	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	2	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	3	

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 Assessments Form – On Validate (DS-01 and DS-02)

```
// Script ID : DS-01, DS-02
// Location  : Assessments form -> On Validate
// Purpose   : Validate entered marks against division by zero and
out-of-range values;
//             derive Percentage and classify Performance Status
using the 50% threshold.

// --- DS-01: Marks Validation Logic ---
if(input.Total_Marks == null || input.Total_Marks <= 0)
{
    alert "Total Marks must be greater than zero.";
    cancel submit;
}
if(input.Marks_Obtained == null || input.Marks_Obtained < 0)
{
    alert "Marks Obtained cannot be negative.";
    cancel submit;
}
if(input.Marks_Obtained > input.Total_Marks)
{
    alert "Marks Obtained cannot exceed Total Marks.";
    cancel submit;
}

// --- DS-02: Percentage and Performance Status Calculation ---
pct = (input.Marks_Obtained * 100.0) / input.Total_Marks;
input.Percentage = pct.round(2);
if(pct < 50)
{
    input.Performance_Status = "Needs Improvement";
}
else
{
    input.Performance_Status = "Good Progress";
}
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

Open a web browser, sign in to Zoho Creator with the authorised Zoho account and open the StudyLens application from the list of applications or via the application URL given in Project at a Glance. The Dashboard is shown first; the other sections are available in the navigation menu. Mobile access has not been tested in the current version.

D.2 Instructions for the Teacher

- Open **Students** and add each student with Student Name, Email and Course; use the Students report to view, edit or search records.
- Open **Subjects** and add the subjects to be assessed (for example, Mathematics).
- Open **Topics**, enter the Topic Name, select its Subject and choose a Difficulty Level (Easy, Medium or Hard).
- Open **Assessments**, select the Student, Subject and Topic, enter Total Marks (greater than 0) and Marks Obtained (between 0 and Total Marks), and submit.
- If an error message appears, correct the marks and submit again.
- Open **Learning Gap Analysis** to view each record's Percentage and Performance Status; records marked "Needs Improvement" are candidates for further review.
- Open the **Dashboard** to view the total number of students, subjects, topics and assessments and the assessment marks chart.

D.3 Instructions for the Application Administrator

- Sign in with the application owner account and open StudyLens in edit mode to modify forms, fields, reports or the Dashboard.
- After any change, repeat the relevant test cases in Table 6.1 before using the application with real data.
- Keep the application private to authorised users and avoid publishing it publicly, because it contains student information.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	ASHVANTH M	Register No.	241501024
Project Title	STUDYLENS – STUDENT LEARNING ANALYZER	Date	

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	ASHVANTH M	Register No.	241501024
Department / Section	Artificial Intelligence and Machine Learning	Date of Viva	
Application URL Verified	Yes / No	Demo Duration	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

Name: _____ Name: _____